Philosophy of Research

Philosophy of Research

Philosophy is the study of general and fundamental problems concerning matters such as existence, knowledge, values, reason, mind, and language

Research can be defined as the search for knowledge or any systematic investigation to establish facts.

Philosophy of Research

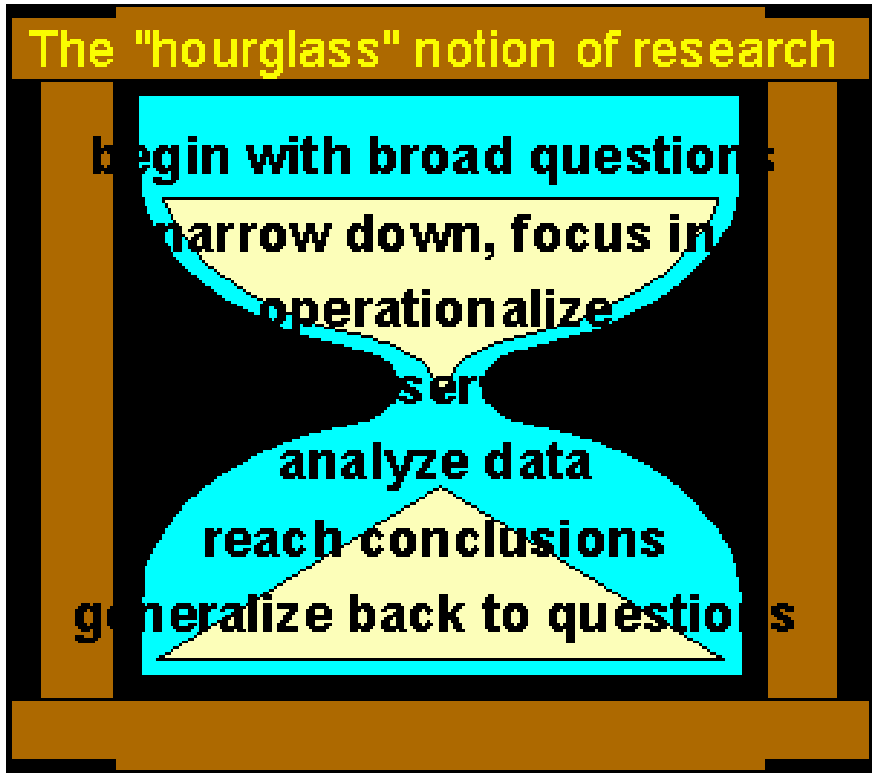
- Why, How, When, What, Research Problem, Objectives, Hypothesis, Analyses, Results, Discussion & Contribution to knowledge

Philosophy of Research

Research Methods and Statistics are **tools**.

Concern is:

- ✓*How* to use them?
- ✓*When* to use them?



Philosophy of Research

STAGE 1:RESEARCH DEFINITION

- ✓Understand the Problem
- ✓Identify Questions
- ✓Refine/Revise Questions

STAGE 2: RESEARCH PLAN/DESIGN

- ✓Choose Design
- ✓Inventory Resources
- ✓Assess Feasibility
- ✓Determine Trade-offs

Philosophy of Research

steps

1. Identified research problem
2. Develop hypothesis
3. Definition –concept & operation (variables)
4. Observation- data collection- survey/experiment/case study etc
5. Analysis data
6. Discussion-interpretation

Research philosophy

‘The term "research philosophy" refers to the foundational approach to understanding the development and essence of knowledge’ Adapted from Saunders *et al*, (2009)

Refers to the “progress of scientific practice based on people’s philosophies and assumptions about the world and the nature of knowledge”

A paradigm is ‘a basic set of beliefs’ that guide action.

Paradigms offer a framework comprising an accepted set of theories, methods, and ways of defining data

Research philosophy

Research philosophy relates to the development of knowledge and the nature of that knowledge

It contains important assumptions about the way in which you view the world

Why does the approach matter?

Whether you take a scientific (positivistic) or Constructivist approach will influence:

- What research questions you ask
- What methods you use to collect your data
- What type of data you collect
- What techniques you use to analyze your data

Positivism: Ideal for testing existing theories, establishing cause-and-effect relationships, and generating generalizable findings.

Constructivism: Well-suited for exploring subjective experiences, understanding social contexts, and uncovering the meanings people ascribe to their experiences.

What AI said:

- Scientific (Positivistic): Focuses on objective, measurable truths and general laws.
- Constructivist: Emphasizes subjective experiences and the co-construction of knowledge.

https://www.youtube.com/watch?v=gmeH2RJ-YpQ

Epistemology, Ontology and Methodology

Epistemology: To do with our beliefs about **how one might discover knowledge about the world**

Ontology: To do with **our assumptions about how the world is made up and the nature of things**

Methodology: To do with the tools and techniques of research

What is Epistemology

The study of our method of acquiring knowledge.

The explanation of how we think.

It answers the question, "How do we know?" I

It encompasses the nature of concepts, the constructing of concepts, the validity of the senses, logical reasoning, as well as thoughts, ideas, memories, emotions, and all things mental.

It is concerned with how our minds are related to reality, and whether these relationships are valid or invalid.

Epistemology

Epistemology comes from the Greek words episteme (knowledge) and it is a branch of philosophy that deals with the ratio, origin and scope of knowledge

It attempts to answer the basic question of what distinguishes true (adequate) knowledge from false (inadequate) knowledge.

It relates to truth and belief.

Importance of Epistemology

- To determine the true from the false by determining a proper method of evaluation.

What are the key elements of a proper Epistemology?

Senses are used to gain information about the world.

Reason is the method of gaining knowledge, and acquiring understanding.

Logic is the method of maintaining consistency within the set of knowledge.

Objectivity is the means of associating knowledge with reality to determine its validity.

Concepts are abstracts of specific details of reality, or of other abstractions. A proper epistemology is a rational epistemology.

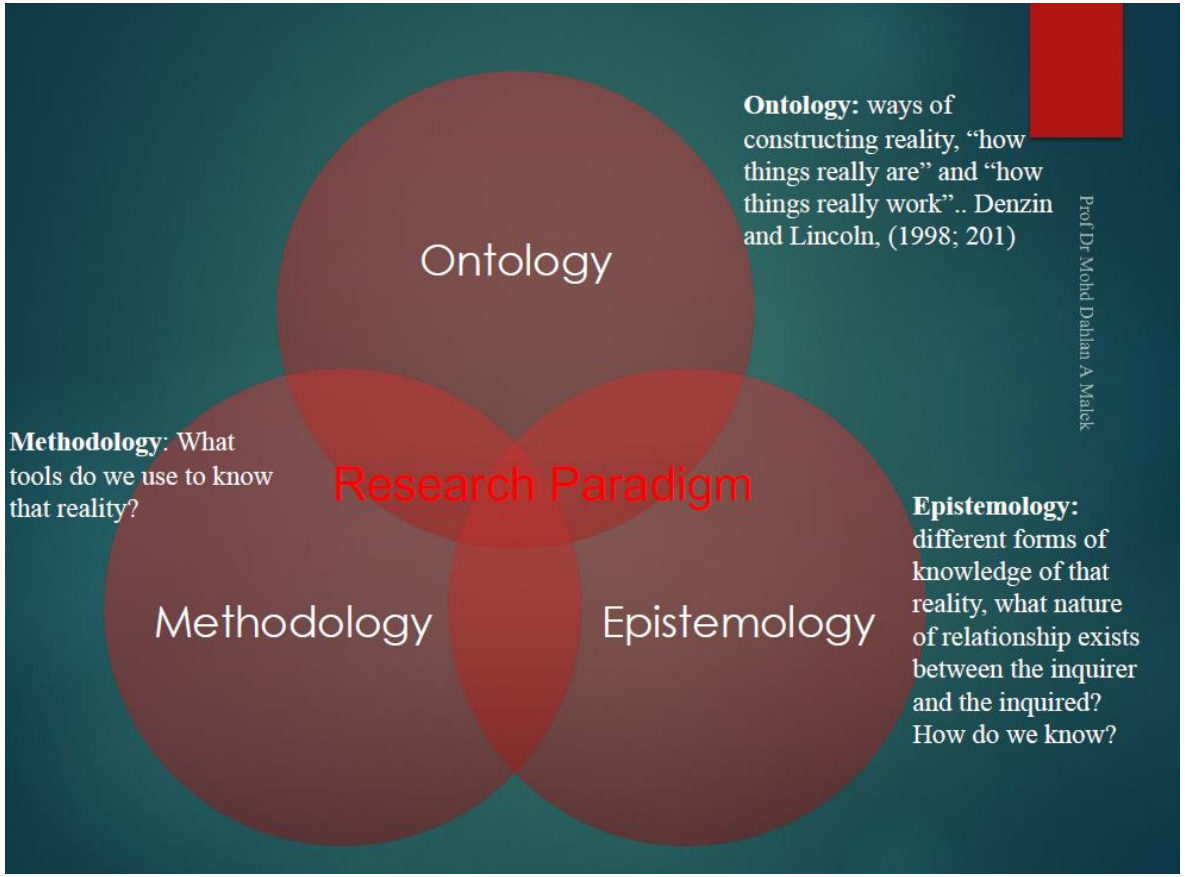
What is an Ontology

Ontology is the study or concern about what kinds of things exist.

Ontology comes from the Greek words **onto which means something that exists**, and **logos which means logical knowledge**.

Gruber defines an ontology as “the specification of conceptualizations, used to help programs and humans share knowledge

The conceptualization is the expressing of knowledge about the world in terms of entities (things, the relationships they hold and the constraints between them)



Ontology: What is reality? (Nature of being)

Epistemology: How do we know? (Nature of knowledge)

Methodology: How do we investigate? (Research process)

Research Paradigms

Positivism - Quantitative *discovery of the laws that govern behavior*

Constructivist - Qualitative *understandings from an insider perspective*

Definition research paradigms

‘A way of examining social phenomenon from which particular understandings of these phenomena can be gained and explanations attempted’ ----Saunders *et al*. (2009)

Positivism - the discovery of laws or backgrounds or factors affecting behavior or phenomenon

Constructivist- researchers acts as ‘social actors’ ; attempt to understand

Selection of Research Paradigm and research

Research paradigm: Positivism

Research Approach: Quantitative

Research Methods: Survey, longitudinal, correlation, experiments

Examples:

- Attitude of students towards online registration system,
- Relationship between students financial position and their academic performance,
- Effect of emotional intelligence on learning effectiveness

Research paradigm: Anti positivism

Research Approach: Qualitative

Research methods: Biographical, phenomenological, ethnological, case-study

Examples:

- A study of autobiography of ……..
- A study of effective management among female managers,
- A case-study of distance learning program

Scientific Research

Provides scientific information and theories

Follows a certain structural process though the step order may vary depending on the subject matter and researcher

The following steps are usually part of the most formal research both basic and applied

- Observations and Formation of the topic
- Hypothesis
- Conceptual definitions
- Operational definition
- Gathering of data
- Analysis of data
- Test, reviving of hypothesis
- Conclusion

Research Methods

The goal of the research process is to produce new knowledge.

This process takes three main forms

- **Exploratory research** which structures and identifies new problems
- **Constructive research** which develops solutions to a problem
- **Empirical research** which tests the feasibility of a solution using empirical evidence.

Level of Research

PhD : Theoretical Gap: Improving the latest theory on research focus; There is contribution of new knowledge.

Master: Empirical gap: Application of tested theoretical framework to solve current problems.

“but not limited”

Research Process

Step 1 - Choose your topic.

Step 2 - Find basic information.

Step 3 - Refine your topic.

Step 4 - Locate and retrieve materials.

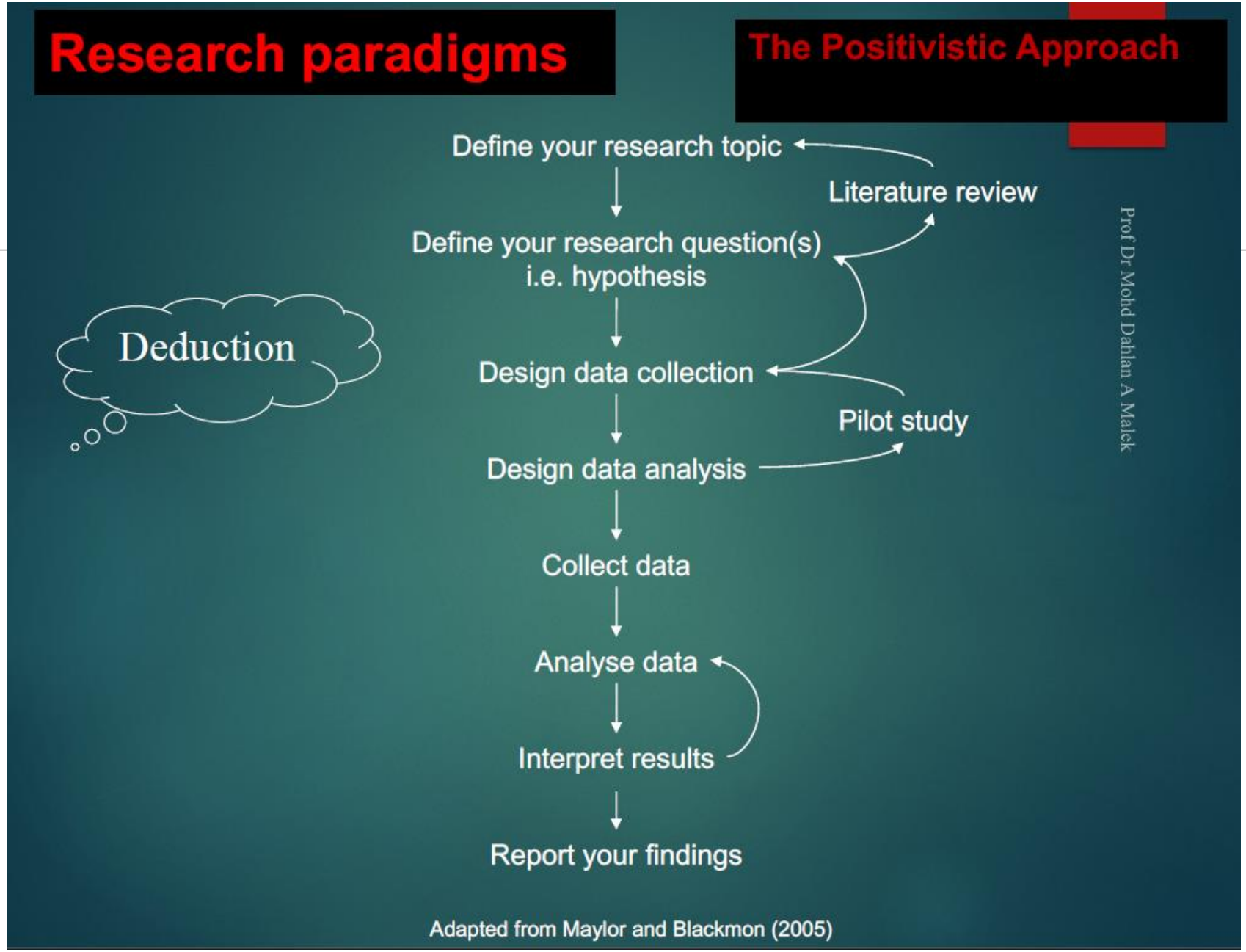
Step 5 - Evaluate relevancies of materials.

Step 6 - Take notes.

Step 7 - Construct your research project. Writing ….

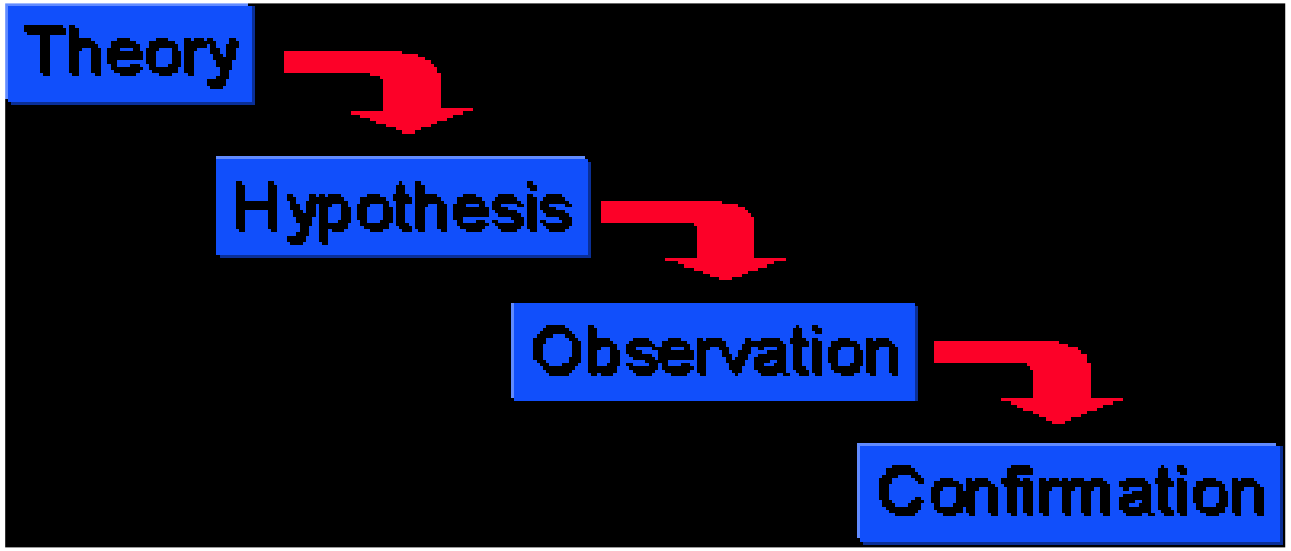
Research paradigms

The Positivistic Approach



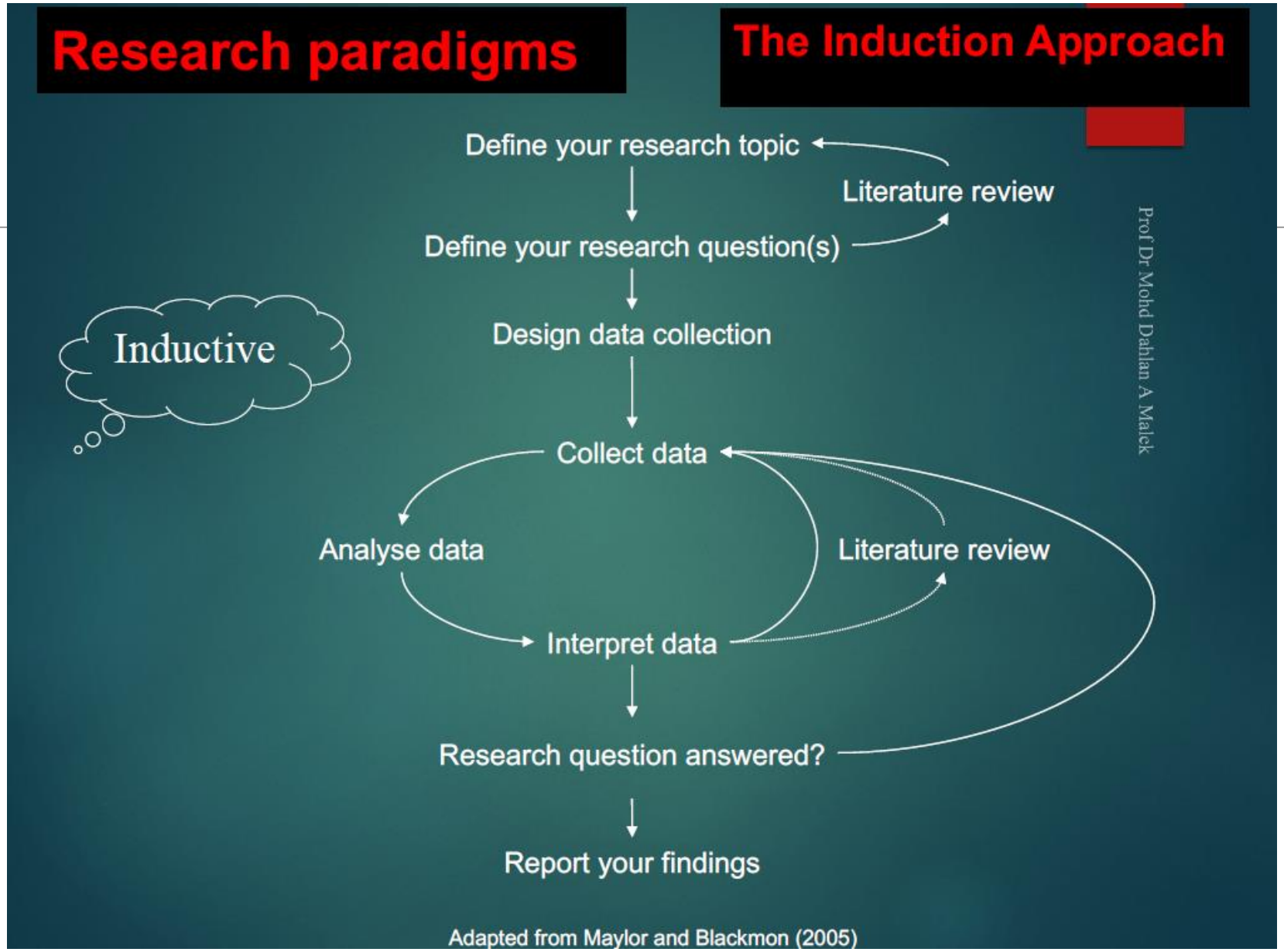
Prof Dr Mohd Dahlan A Malek

Deduction



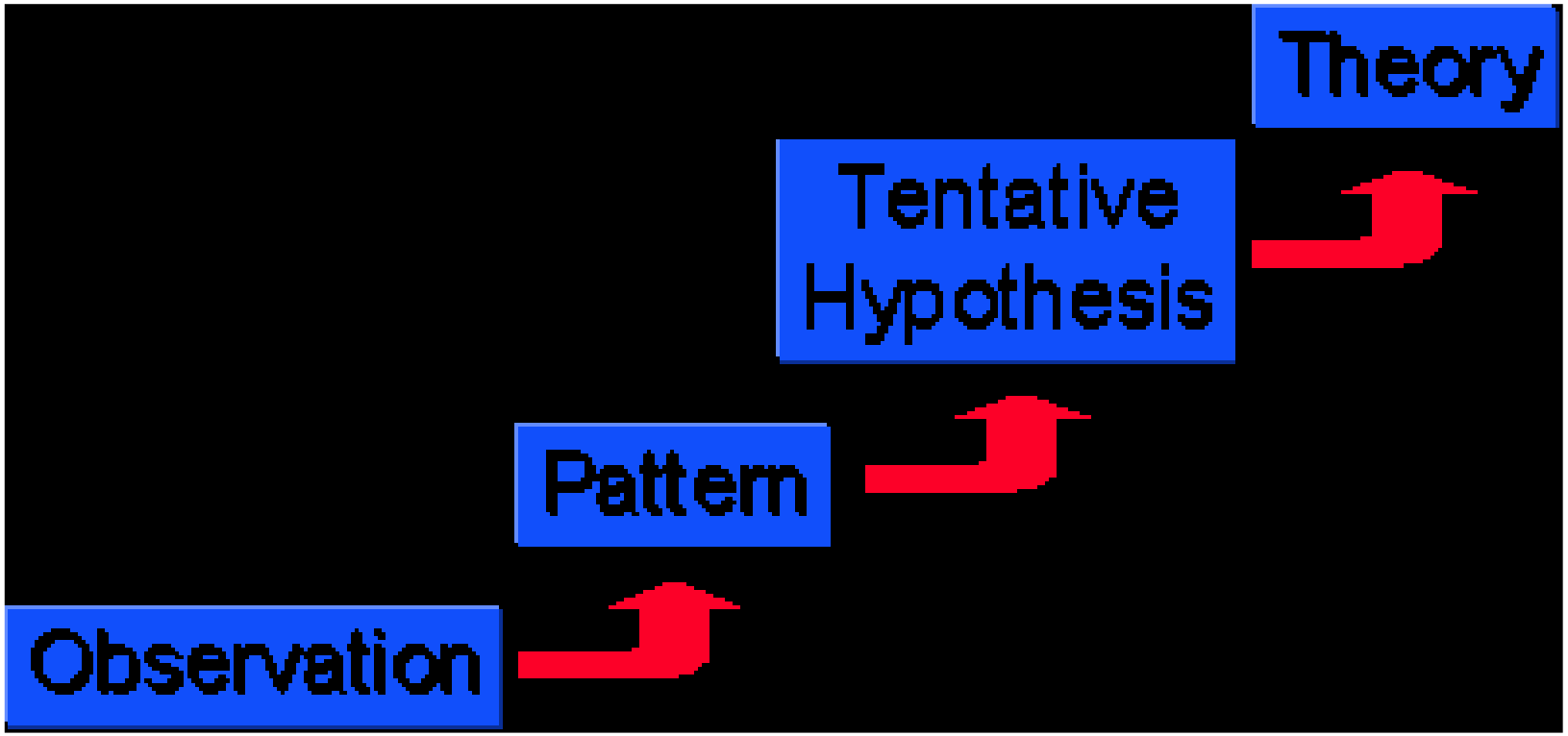
Research paradigms

The Induction Approach



Adapted from Maylor and Blackmon (2005)

Induction



Choosing your research approach

The right choice of approach helps you to:

Make a more informed decision about the research design

Think about which strategies will work for your research topic

Adapt your design to cater for any constraints

Adapted from Easterby-Smith *et al*. (2008)

Things worth considering

The nature of the research topic

The time available

The extent of risk

The research audience – managers

Deduction emphasises	Induction emphasises
• scientific principles • moving from theory to data • the need to explain causal relationships between variables • the collection of quantitative data • the application of controls to ensure validity of data • the operationalisation of concepts to ensure clarity of definition • a highly structured approach • researcher independence of what is being researched • the necessity to select samples of sufficient size in order to generalise conclusions	• gaining an understanding of the meanings humans attach to events • a close understanding of the research context • the collection of qualitative data • a more flexible structure to permit changes of research emphasis as the research progresses • a realisation that the researcher is part of the research process • less concern with the need to generalise

Comparing Approaches

Characteristic	Positivism	Constructivism
Questions that can be answered	What? How much?	Why? How?
Associated methods	Survey, Experiment	Direct observation, Interviews, Participant observation
Data type	Predominantly numbers	Predominantly words
Finding	Measure	Meaning

Adapted from Maylor and Blackmon (2005)

Combining research approaches

MULTIPLE-METHODS APPROACH

- **Triangulation**: Applying 2 or more *dissimilar* measures and/or methods (research strategies) to investigating a certain problem.
- Why do it:
  - increased confidence in
  - Findings

Multiple methods can also be used within a single approach:

- allows exploiting the strengths & weaknesses of complementary methods.

Example:

- One approach (survey method), but mail questionnaire to probability sample, & face-to-face interviews on smaller sample of non-respondents, to estimate non-response bias.
- experimental designs in survey research;
- Use of archival records to identify groups for field research …